

Code-Layout, Readability and Reusability

Date	15 March 2024 June 2026
Team ID	XXXXXX EduGenie - Google Gemini Powered Learning Assistant
Project Name	XXXXXX EduGenie - Google Gemini Powered Learning Assistant
Maximum Marks	5 Marks

Code-Layout, Readability and Reusability:

This document evaluates the quality of the codebase in terms of its structure, readability, and reusability. Well-organized code with proper naming conventions, comments, and modular design ensures that the project is maintainable and can be extended or reused in future development.

Code Layout Checklist:

S.No	Code Quality Parameter	Description	Followed (Yes / No / Partial)	Remarks
1	Consistent Indentation	Uniform spacing/tabs used throughout the code		
2	Proper File Structure	Files and folders are logically organized	ed	
3	Meaningful Variable Names <code>call_gemini_with_retry()</code>	Variables reflect their purpose clearly Python (utils/gemini_service.py) Used indirectly by every get_*() response in High gemini_service_builder ice.py ice.py		
4	Function / Method Names <code>log_query()</code> <code>log_ai_response()</code>	Functions are descriptively named Python (utils/db_manager.py) Called from learning_pa learning_pa	every router (qna, explanation, quiz, th) and main.py th) and main.py	summary, High
5	Code Comments <code>get_base_context()</code>	Inline and block comments explain log Shared by a	HTML page routes to pass gemini C	configured to Medium
6	Modular Design <code>init_local_t5()</code> <code>clean_json_block()</code>	Python (explanation_module.py) Lazily loaded Code is split into reusable functions/m	d T5 pipeline, reusable for future local dules down json fences from LLM output	l-model Medium fore JSON
7	No Redundant Code	Duplicate or unused code is removed parsing parsing		Medium
8	Error Handling	Exceptions and errors are handled gracefully	cefully	

Reusable Components / Modules:

S.No	Component / Module Name	Language / Technology	Where Reused	Reusability Level (High / Medium / Low)
1				

2				
3				
4				
5				

Overall Code Quality Assessment:

Aspect	Rating (1-5)	Comments
Code Layout & Structure		
Readability		
Reusability		
Documentation / Comments		
Overall Score		

call_gemini_with_retry()	Python (utils/gemini_service.py)	Used indirectly by every get_*(*) response builder in gemini_service.py gemini_service.py	High
log_query() / log_ai_response()	Python (utils/db_manager.py)	Called from every router (qna, explanation, quiz, summary, learning_path) and main.py learning_path) and main.py	High
get_base_context()	Python (main.py)	Shared by all HTML page routes to pass gemini_configured to templates templates	Medium
init_local_t5()	Python (explanation_module.py)	Lazily loaded T5 pipeline, reusable for future local-model features features	Medium
clean_json_block()	Python (quiz_module.py)	Strips markdown json fences from LLM output before JSON parsing parsing	Medium